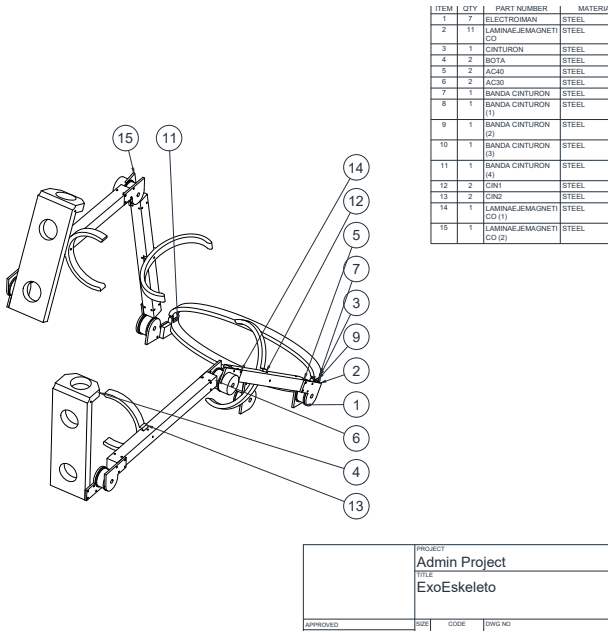


PRELIMINARY ENGINEERING REPORT

Motor and Joint-Torque Analysis

ExoEskeleto Drawing v1



HIP
120–150 N m
peak output, each side

KNEE
80–110 N m
peak output, each side

ANKLE
120–150 N m
peak output, each side

Source drawing

ExoEskeleto Drawing v1.pdf, 17 sheets

Drawing date

13 July 2026

Reference user

80 kg

Report status

Preliminary sizing; not approved for human testing

Prepared from the geometry and bill of materials supplied in the source drawing.

Scope and safety limitation. This report is a first-pass actuator-sizing study based on the drawing and explicitly stated assumptions. It is not a final design approval or safety certification. Complete CAD mass-property extraction, multibody simulation, structural FEA, brake validation, thermal/electrical analysis, and controlled bench testing are required before testing on a person.

Contents

1	Executive summary	1
2	Drawing basis and observed geometry	2
2.1	Information established by the drawing	2
2.2	Information not supplied	2
3	Reference design case	2
4	Static gravity calculations	3
4.1	Hip: one leg held horizontal	3
4.2	Knee: shank and foot held horizontal	3
4.3	Ankle: ground-reaction load governs	3
5	Lagrange-Euler torque and power calculation	4
5.1	Three-link planar model	4
5.2	Numerical input state	4
5.3	Calculated torque and joint power	5
5.4	Illustrative gearbox mapping	5
6	Motor and transmission sizing	6
6.1	Torque and ratio relationship	6
6.2	Recommended architecture	6
6.3	Scaling and partial assistance	6
7	Electromagnet feasibility check	7
8	Structural and integration observations	7
9	Data required for final sizing	8
10	Conclusion	8

1 Executive summary

The 17-sheet drawing establishes the overall exoskeleton arrangement and principal link geometry. It does not define rotary motors, gearboxes, rated joint torques, component mass properties, actuator

moment arms, gait profiles, or an assistance target. The bill of materials instead identifies seven electromagnets and eleven magnetic laminations or plates.

For a reference 80 kg user and a full lower-limb support envelope, the recommended preliminary output-side ratings are shown in Table 1. Six powered flexion/extension joints are implied if both hip, knee, and ankle joints are actuated.

Applying the Lagrange-Euler equation to the report's three-link reference model gives, for the stated horizontal dynamic example, 60.21 N m and 90.31 W at the hip, 17.99 N m and 35.98 W at the knee, and 2.54 N m and 6.35 W at the ankle. These are unfactored swing-phase values for human-segment mass only; they exclude exoskeleton mass and ground-reaction force.

Table 1: Preliminary joint-output design envelope, each side.

Joint	Continuous torque	Peak torque	Output speed	Approximate actuator class
Hip flexion/extension	45–60 N m	120–150 N m	0–3 rad/s	300–500 W BLDC with low-backlash reducer
Knee flexion/extension	30–45 N m	80–110 N m	0–4 rad/s	250–400 W BLDC with low-backlash reducer
Ankle plantar/dorsiflexion	45–60 N m	120–150 N m	0–5 rad/s	350–600 W BLDC with reducer or remote transmission

Primary design finding. The electromagnets should be treated as auxiliary locks, clutches, or mode-selection devices unless their force-versus-gap and friction performance are demonstrated. They should not be assumed to replace torque-controlled rotary actuators or certified fail-safe brakes.

2 Drawing basis and observed geometry

2.1 Information established by the drawing

- Overall envelope: approximately 1009 mm high, 1009 mm long, and 662 mm wide (sheet 2).
- Principal long members: approximately 400–420 mm on the detail sheets.
- Several circular joint/interface details: approximately 50 mm outside diameter with 10 mm bores.
- Bill of materials: seven ELECTROIMAN units and eleven magnetic laminations or plates.
- Listed structural parts are identified as steel.
- The assembly appears to comprise bilateral lower-limb links connected to a waist ring and foot/boot interfaces.

2.2 Information not supplied

The drawing does not provide maximum user mass, payload, part masses, centres of mass, gait cycle, required assistance percentage, motor torque-speed curves, gearbox ratios or efficiencies, bearing friction, joint range, structural steel grade, weld specifications, or electromagnet force data. These omissions prevent an exact CAD-derived actuator calculation.

3 Reference design case

The assumptions in Table 2 provide a transparent first-pass sizing case. Segment masses are anthropometric approximations and exclude exoskeleton hardware.

Table 2: Reference assumptions used in the calculations.

Parameter	Assumption
User mass	80 kg
Thigh length	0.42 m
Shank length	0.40 m
Foot COM distance from ankle	0.13 m
One thigh mass	10.0% body mass = 8.00 kg
One shank mass	4.65% body mass = 3.72 kg
One foot mass	1.45% body mass = 1.16 kg
Thigh and shank COM location	43.3% of segment length from proximal joint
Gravity, g	9.81 m/s ²
Dynamic multiplier	2.0
Engineering factor after dynamics	1.5

The dynamic and engineering factors are included to establish an actuator selection envelope. Final load cases must instead use measured or simulated joint torque histories and include every moving exoskeleton component.

4 Static gravity calculations

For a point mass, the gravity moment about a joint is

$$T = mgr, \quad (1)$$

where r is the horizontal distance from the joint axis. Gravity torque is largest when the segment is approximately horizontal.

4.1 Hip: one leg held horizontal

$$T_{\text{hip}} = g[m_t(0.433L_t) + m_s(L_t + 0.433L_s) + m_f(L_t + L_s + r_f)]. \quad (2)$$

For the 80 kg reference user,

$$T_{\text{hip}} = 46.7 \text{ N m} \quad (3)$$

for limb self-weight alone. This does not include the exoskeleton links, actuator masses, payload, or ground-reaction effects.

4.2 Knee: shank and foot held horizontal

$$T_{\text{knee}} = g[m_s(0.433L_s) + m_f(L_s + r_f)], \quad (4)$$

giving

$$T_{\text{knee}} = 12.3 \text{ N m}. \quad (5)$$

4.3 Ankle: ground-reaction load governs

Foot self-weight produces only about 1.5 N m. This is not the governing ankle case. During standing and walking, ground-reaction force acts at a horizontal offset from the ankle. For the complete 80 kg

body load and a 50–120 mm centre-of-pressure offset,

$$T_{\text{ankle}} = (80)(9.81)(0.05 \text{ to } 0.12) \quad (6)$$

$$= \boxed{39 \text{ to } 94 \text{ N m}}. \quad (7)$$

In symmetric double support, each side may carry approximately half the load. In single support, one side can approach the full value and can exceed it dynamically.

5 Lagrange-Euler torque and power calculation

5.1 Three-link planar model

Let

$$\mathbf{q} = [q_1 \ q_2 \ q_3]^T \quad (8)$$

contain the relative hip, knee, and ankle angles. The pelvis is treated as a fixed base; q_1 is measured from the global horizontal, while q_2 and q_3 are relative angles. Positive rotation is counter-clockwise.

For the three moving segments, the kinetic and potential energies are

$$K = \frac{1}{2} \sum_{i=1}^3 \left(m_i \dot{\mathbf{r}}_i^T \dot{\mathbf{r}}_i + I_i \omega_i^2 \right), \quad (9)$$

$$V = \sum_{i=1}^3 m_i g y_i, \quad (10)$$

and the Lagrangian is $\mathcal{L} = K - V$. Applying the requested equation at each joint,

$$\tau_i = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i}, \quad (11)$$

gives the manipulator form

$$\boxed{\boldsymbol{\tau}_{\text{act}} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) - \mathbf{J}_f^T \mathbf{F}_{\text{ext}}}. \quad (12)$$

The final term is essential in stance: $\mathbf{F}_{\text{ext}}$ is the measured ground-reaction force at the foot and $\mathbf{J}_f$ is the foot Jacobian. It is set to zero in the numerical swing example because no force history is supplied.

Segment inertias are estimated using $I_i = m_i(k_i L_i)^2$. The radius-of-gyration fractions for thigh, shank, and foot are 0.323, 0.302, and 0.475, respectively. The calculation includes human segment mass only.

5.2 Numerical input state

The selected instant is deliberately explicit so that the result can be reproduced or replaced with measured data.

Table 3: State used for the illustrative Lagrange-Euler calculation.

Quantity	Hip	Knee	Ankle
q_i	0°	0°	0°
$\dot{q}_i$	1.5 rad/s	2.0 rad/s	2.5 rad/s
$\ddot{q}_i$	3.0 rad/s ²	4.0 rad/s ²	5.0 rad/s ²
External foot force	$\mathbf{F}_{\text{ext}} = [0 \ 0]^T \text{ N}$		

At this collinear horizontal configuration,

$$\mathbf{M} = \begin{bmatrix} 2.8397 & 1.0382 & 0.1610 \\ 1.0382 & 0.5094 & 0.0976 \\ 0.1610 & 0.0976 & 0.0373 \end{bmatrix} \text{ kg m}^2, \quad (13)$$

$$\mathbf{G} = \begin{bmatrix} 46.7308 \\ 12.3518 \\ 1.4793 \end{bmatrix} \text{ N m}, \quad \mathbf{C}\dot{\mathbf{q}} = \mathbf{0}. \quad (14)$$

The Coriolis/centrifugal vector is zero at this particular collinear state because the required sine coupling terms vanish; it is generally non-zero at other postures.

5.3 Calculated torque and joint power

Instantaneous joint mechanical power is

$$P_i = \tau_i \dot{q}_i. \quad (15)$$

Positive power drives the motion; negative power indicates braking or potential regeneration.

Table 4: Unfactored dynamic result for the state in Table 3.

Joint	Inertial (N m)	Gravity (N m)	Total torque (N m)	Joint power (W)
Hip	13.477	46.731	60.208	90.311
Knee	5.640	12.352	17.992	35.985
Ankle	1.060	1.479	2.539	6.348
Total				132.644

For example, the first-joint calculation is

$$\tau_1 = 2.8397(3) + 1.0382(4) + 0.1610(5) + 46.7308 \quad (16)$$

$$= \boxed{60.208 \text{ N m}}, \quad (17)$$

$$P_1 = \tau_1 \dot{q}_1 = 60.208(1.5) = \boxed{90.311 \text{ W}}. \quad (18)$$

If a separate 1.5 engineering factor is applied after this trajectory calculation, the resulting provisional values are 90.31, 26.99, and 3.81 N m for hip, knee, and ankle. The ankle value remains unsuitable for stance sizing because the ground-reaction term is absent.

5.4 Illustrative gearbox mapping

Using 80:1 hip and knee reducers, a 60:1 ankle reducer, and 75% gearbox efficiency gives Table 5. Shaft power is joint power divided by gearbox efficiency; electrical input will be higher again after motor and drive losses.

Table 5: Motor-side result for the illustrative swing state.

Joint	Ratio	Motor torque (N m)	Motor speed (rpm)	Shaft power (W)
Hip	80:1	1.003	1146	120.4
Knee	80:1	0.300	1528	48.0
Ankle	60:1	0.056	1432	8.5

As a conservative instantaneous envelope, multiplying the preliminary peak torque and maximum speed in Table 1 gives 450 W at the hip, 440 W at the knee, and 750 W at the ankle. Torque and speed peaks normally occur at different times, so these products are upper bounds rather than continuous motor ratings. Final continuous power must be based on cycle RMS torque, measured speed histories, gearbox efficiency maps, and thermal limits.

6 Motor and transmission sizing

6.1 Torque and ratio relationship

For gearbox ratio N and transmission efficiency η , required motor torque is

$$T_{\text{motor}} = \frac{T_{\text{joint}}}{N\eta}. \quad (19)$$

For a 120 N m peak joint, an 80:1 ratio, and 75% efficiency,

$$T_{\text{motor,peak}} = \frac{120}{80(0.75)} = \boxed{2.0 \text{ N m}}. \quad (20)$$

At 60:1 and 80% efficiency, the same joint requires 2.5 N m motor peak torque. A reduction range near 50:1–100:1 is a reasonable starting point, but the final ratio must satisfy joint torque, joint speed, thermal duty, backlash, and backdrivability requirements simultaneously.

6.2 Recommended architecture

- 48 V BLDC or PMSM actuators to limit current and cable mass.
- Absolute joint encoder plus motor encoder.
- Low-backlash harmonic, cycloidal, or well-supported planetary reducer.
- Torque sensing through a strain element or series-elastic element.
- Normally engaged mechanical brake at each load-bearing joint.
- Mechanical hard stops independent of software.

High reduction improves torque density but reduces backdrivability and increases reflected inertia. Torque sensing and mechanical compliance are strongly preferred for a wearable device. Distal motor mass also increases swing torque; proximal knee actuation and remote ankle transmission should be evaluated.

6.3 Scaling and partial assistance

For a user mass M different from 80 kg, a first approximation is

$$T_{\text{new}} = T_{\text{table}} \left(\frac{M}{80} \right) + T_{\text{exo+payload}}. \quad (21)$$

For partial assistance, multiply the biological joint-torque target by the desired assistance fraction. For example, 40% assistance at a 150 N m ankle peak requires approximately 60 N m actuator output before transmission losses and friction are added.

7 Electromagnet feasibility check

If an electromagnet at a nominal 50 mm diameter interface is expected to resist joint torque through friction at an effective radius of approximately 25 mm, the required tangential force is

$$F_t = \frac{T}{r}. \quad (22)$$

Table 6: Tangential force implied at a 25 mm effective radius.

Joint torque	Tangential force
60 N m	2400 N
120 N m	4800 N
150 N m	6000 N

Friction torque is $T = \mu F_n r$. With a nominal dry friction coefficient of 0.3, holding 120 N m at a 25 mm radius requires

$$F_n = \frac{120}{0.3(0.025)} = \boxed{16,000 \text{ N}} \quad (23)$$

of magnetic normal force. Air gaps, surface contamination, coil heating, and loss of electrical power further reduce dependable holding capacity.

Design implication. Do not use the drawn electromagnets as the sole load-bearing joint brakes without a demonstrated force/torque test. If used for locking or mode selection, add a mechanically positive, normally locked feature so that loss of power cannot release the user.

8 Structural and integration observations

- Several details show 10 mm bores. A pin at this interface requires combined shear, bearing, bending, fatigue, and retention analysis; bore diameter alone is insufficient.
- Long 400–420 mm links and offset cuffs introduce bending and torsion in addition to in-plane joint torque.
- The BOM states steel but does not provide grade, complete thickness data, heat treatment, weld details, or allowable stress.
- Joint axes must align with the user’s anatomical hip, knee, and ankle axes. Misalignment can create harmful cuff forces even when motor torque is within its limit.
- Knee motors should preferably be placed proximally. The ankle may benefit from cable, Bowden, belt, or linkage transmission to reduce distal mass.
- Motor torque must not be the only protection against collapse during power loss.

9 Data required for final sizing

1. Maximum user mass and carried payload.
2. Intended function: rehabilitation assistance, full weight support, sit-to-stand, walking, or static locking.
3. Required degrees of freedom, angular limits, and mechanical stop positions.
4. Target gait speed and joint angle, velocity, and acceleration profiles versus time.
5. CAD mass and centre of mass for each link, motor, gearbox, battery, and attachment.
6. Electromagnet model, force-versus-gap curve, coil power, friction material, effective radius, and fail-safe behaviour.
7. Gearbox ratio, efficiency, backlash, rated output moment, shock rating, and life.
8. Ground-reaction-force cases and required assistance percentage.

10 Conclusion

For an 80 kg reference user and full lower-limb support, size each side initially for 120–150 N m peak at the hip, 80–110 N m peak at the knee, and 120–150 N m peak at the ankle, with the continuous ratings in Table 1. These are output-side design-envelope values, not motor-shaft values.

The seven electromagnets identified in the drawing are best treated as auxiliary locks or clutches. They are not substitutes for torque-controlled rotary motors or validated fail-safe brakes. Human testing should not begin until the missing design inputs are resolved and the system has passed simulation, structural verification, brake testing, thermal analysis, electrical safety review, and controlled bench trials.

Next engineering release. Replace the reference anthropometric model with CAD mass properties and measured motion/load cases, then issue a joint-by-joint motor, reducer, brake, bearing, pin, and controller specification.